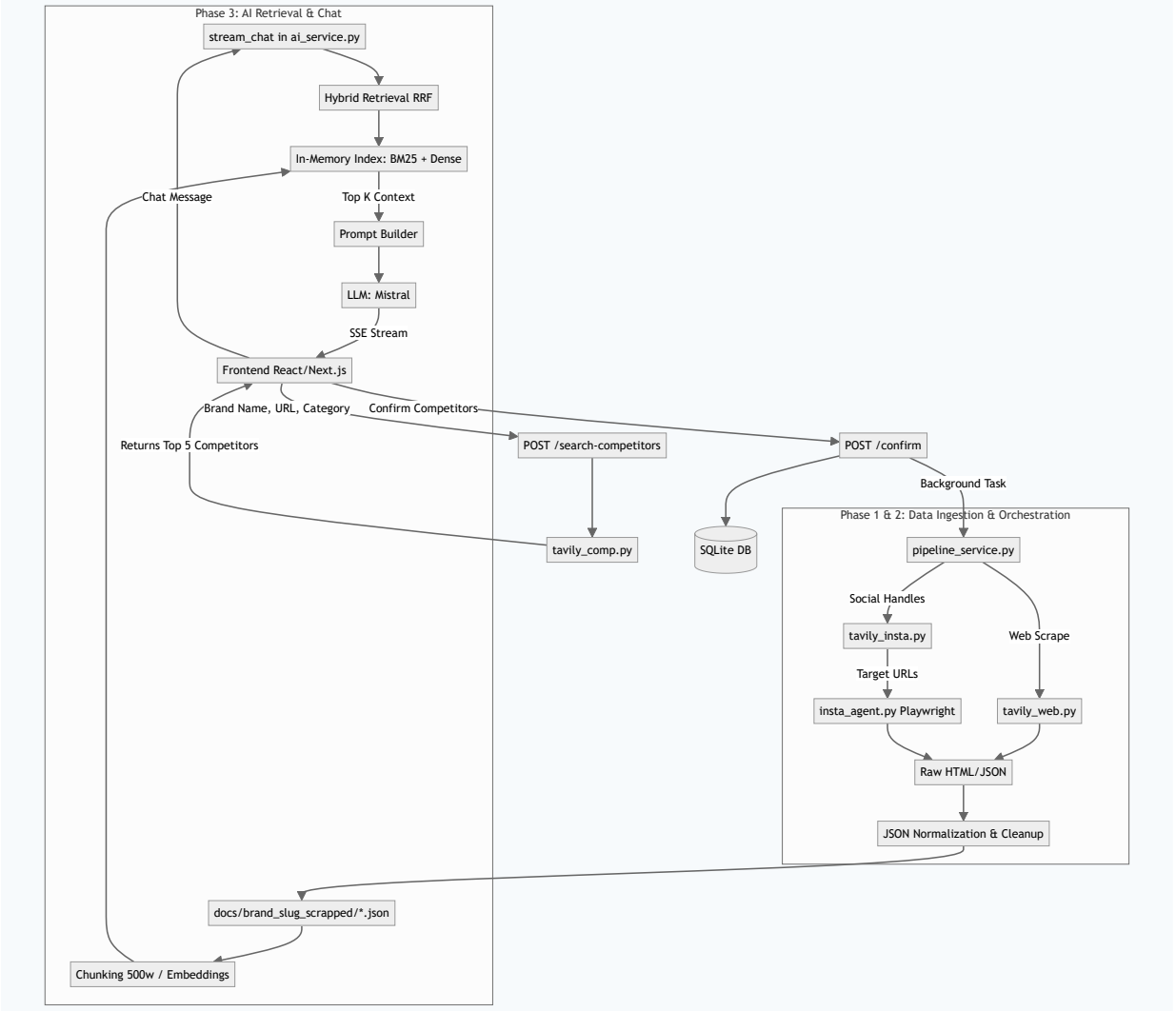
StratAI Architecture & Technical Documentation

This document provides a comprehensive overview of the StratAI platform's architecture, data flow pipelines, user journeys, and potential avenues for future intelligence enhancements and configuration.

1. Phase-Wise Architecture & Data Flow

The StratAI architecture is divided into three primary phases: `Data Ingestion`, `Data Processing & Orchestration`, and `Retrieval-Augmented Generation (RAG) & Chat`.

Architecture & Data Flow Diagram



Phase 1: Data Ingestion (Scraping & Discovery)

This phase handles fetching raw external data about the user's brand and their competitors. It relies on standalone Python scripts located in the `web_scrapper` and `social_scrapper` directories.

- **Competitor Discovery (tavily_comp.py)**: Uses the Tavily API to query for direct market competitors.
- **Deep Web Research (tavily_web.py)**: Uses Tavily Advanced Search to perform a deep dive into 4 specific areas.
- **Social Handle Discovery (tavily_insta.py)**: Automatically finds official Instagram handles.
- **Instagram Scraper (insta_agent.py)**: Uses Playwright to navigate Instagram, scraping recent posts, captions, likes, and comments.

Phase 2: Data Processing & Orchestration (pipeline_service.py)

When a user initiates research, the FastAPI backend kicks off an asynchronous background task.

1. **Initialization**: Creates a `docs/{brand_slug}_scrapped` directory.
2. **Execution**: Invokes web scraping, deep web research, social handle discovery, and spawns the Playwright `insta_agent`.
3. **Normalization**: Converts all output files (stripping CSS/JS and replacing emojis) into standardized JSON formats.
4. **Completion**: Updates the SQLite database.

Phase 3: AI Retrieval & Chat (ai_service.py)

This phase handles answering user queries using the ingested data.

1. **Indexing**: Chunks the text and embeds using `all-MiniLM-L6-v2`.
2. **Hybrid Retrieval**: Uses Cosine similarity (Dense) and BM25 (Sparse) fused via RRF.
3. **Generation**: Context is injected into a strict System Prompt and passed to the LLM (`mistral-large-latest`).

2. User Journey

1. **Onboarding**: User signs in and initiates a "New Brand Research" workflow.
2. **Brand Input**: User enters Brand Name, Website URL, and Industry Category.
3. **Competitor Review**: AI suggests top 5 competitors. User reviews, edits, and clicks "Confirm & Start".
4. **Background Processing**: User drops into the Chat UI while background pipeline scrapes web and Instagram.
5. **Strategic Consultation**: User asks strategic questions (e.g., "What were the likes on my last 5 posts?").
6. **Insight Delivery**: StratAI delivers concise, data-backed insights.

3. Future Intelligence Enhancements

- **Smart Competitor Curation**: Analyze *why* they are a competitor.
- **Automated Sentiment Analysis**: Score Instagram comments (Positive/Neutral/Negative).
- **Automated SWOT Analysis**: Generate a complete SWOT analysis at the end of the pipeline.
- **Query Routing**: Route intents (Visual/Chart, Numeric Analysis, General Strategy) to specialized agents.